

Corporate Share Repurchase Forecasting for Cross-Sectional Stock Selection

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Abstract

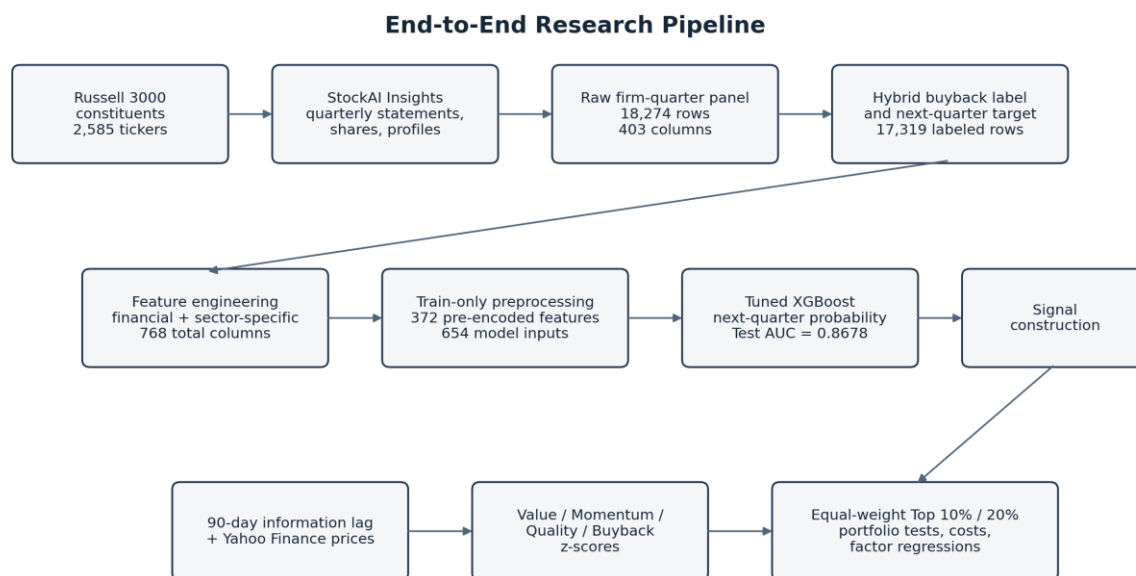
This investigation builds an end-to-end machine learning pipeline to forecast whether a public company will repurchase shares in the following quarter and then asks a separate investment question of whether a high predicted probability of repurchase improves cross-sectional stock selection. Starting from a Russell 3000 constituent file, the project assembled a custom firm-quarter panel from quarterly financial statements, share data, company profiles, and market prices. The resulting labeled sample contains 17,319 firm-quarter observations across 954 public companies and was expanded through general and sector-specific feature engineering. A tuned XGBoost classifier achieved a 0.8678 ROC-AUC on the held-out test set, compared with 0.7763 for a one-feature persistence baseline. In contrast, a fundamentals-only XGBoost specification that removed direct buyback and share-change predictors retained a 0.8611 test AUC, indicating that the forecast is not solely a mechanical extrapolation of recent repurchase behavior. The second stage converts model probabilities into quarterly stock-selection signals and compares buyback-only portfolios with Value-Momentum-Quality (VMQ) composites. In the 15-quarter investable backtest, buyback only portfolios lagged both an equal-weight benchmark and VMQ portfolios, while adding the buyback score to VMQ reduced returns. At the Top 10% cutoff, stocks introduced by the buyback signal underperformed the VMQ names they replaced by 7.74 percentage points per quarter on average (HC3 $p = 0.045$). The result is therefore a distinction between corporate-action predictability and asset-pricing usefulness: the model predicts repurchase behavior well, but the predicted behavior did not operate as a standalone return factor in this limited sample.

1. Research Question and Contribution

Prior research identifies several motivations for stock buybacks, including returning excess cash to shareholders, correcting perceived undervaluation, supporting stock prices and liquidity, and conveying information to the market; repurchase decisions are also closely connected to firms' broader capital structure and compensation policies (Chen & Obizhaeva, 2022). Share repurchases are an observable corporate financing decision with a useful predictive structure as they are discrete enough to frame as a classification problem arising from a broad combination of profitability, cash generation, leverage, valuation, capital structure, and recent payout behavior. This project attempts to answer two linked but deliberately distinct questions. First, can next-quarter share repurchases be predicted from information available in company financial statements and market data? Second, if they can be predicted, does the resulting probability contain incremental information for stock selection beyond conventional Value, Momentum, and Quality signals?

This distinction matters since successful prediction of a corporate action does not imply successful prediction of returns. A model can learn persistent balance sheet or financing patterns that accurately anticipate a repurchase while the market already prices those patterns, or repurchases occur for reasons that do not create abnormal future returns. Treating the classification and portfolio problems separately prevents predictive accuracy from being mistaken for investment alpha.

Essentially, from the experiment, the classification problem is strong; the investment signal is weak. The final XGBoost model reached 0.8678 test ROC-AUC, but buyback-ranked portfolios underperformed the equal-weight universe and adding the buyback score to VMQ reduced the composite portfolio return in this backtest.

Figure 1*End-to-End Research Architecture*

The project separates corporate-action prediction from the subsequent question of whether the prediction has cross-sectional return value.

Note. The workflow proceeds from universe construction through corporate-action prediction and portfolio testing. Counts are those produced by the project notebooks. Image generated by GPT-5.6 Sol as it was suggested by the AI.

The contribution of the project is primarily methodological. Rather than starting from a pre-cleaned machine learning dataset, the analysis constructs the panel, defines the target, engineers cross-sector accounting features, applies train-only preprocessing, compares several model classes, tests a persistence baseline, removes direct buyback-history variables as a robustness specification, and then converts the model output into an explicitly timed portfolio signal. The negative portfolio result is retained rather than optimized away as it is part of the research conclusion.

2. Data Architecture and Buyback Label

Universe and Firm-Quarter Panel Construction

The starting universe was a cleaned Russell 3000 constituent file containing 2,585 ticker symbols. Ticker conventions were normalized for compatibility across data sources. Quarterly income statements, balance sheets, cash-flow statements, share information, and company profiles were requested through the StockAI Insights API and merged by company and reporting period. The API build produced usable panels for 955 tickers and empty or failed panels for 1,630 tickers. Before target shifting, the resulting raw panel contained 18,274 firm-quarter observations and 403 columns with no duplicate ticker-period keys.

Table 1

Dataset Construction Summary

Build stage	Count	Interpretation
Constituent tickers attempted	2,585	Cleaned Russell 3000 symbol list
Tickers with panels built	955	At least one usable quarterly panel
Raw firm-quarter observations	18,274	Before next-quarter target shift
Raw columns	403	Statement, share, profile, and derived fields
Labeled observations	17,319	Rows with observable next-quarter outcome
Labeled unique tickers	954	Issuers retained after shifting the label
Positive-class rate	42.62%	Next-quarter buyback = 1

The data is intentionally a panel rather than a single cross section with each observation representing a company-quarter, permitting lagged and growth features, current-quarter behavior, and subsequent-quarter labels but also creating repeated observations for the same issuer (this point is revisited in the limitations section because the model split is stratified at the row level rather than grouped by issuer or time).

Hybrid Buyback Label

The current-quarter buyback indicator combines the provider buyback flag with a share-count rule. A quarter is labeled as a buyback when either the provider reports a buyback or the reported share-count percentage change is less than or equal to -1.25%. The prediction target is then generated by shifting that indicator forward one quarter within each ticker. The features $X(i,t)$ are used to predict $Y(i,t+1)$ where Y equals one when the next quarter satisfies the hybrid repurchase rule:

$$Y(i,t+1) = 1 \{ \text{provider buyback flag}(i,t+1) = \text{True or share-count change}(i,t+1) \leq -1.25\% \}$$

The final labeled build contained 9,938 non-buyback observations and 7,381 buyback observations in the build output, corresponding to a 42.62% positive rate. Exploratory work on the final feature file reported the class distribution as approximately 57.4% versus 42.6%. The analysis therefore treats the problem as somewhat balanced.

Price Data and later Return Mapping

Quarterly price fields were attached to the firm-quarter panel for feature engineering. The later investment backtest uses a separate, stricter daily-price process from Yahoo Finance through yfinance. This separation is important as the classifier can be evaluated without requiring a complete future return history, while the portfolio stage can impose common trading dates and remove observations that cannot be priced consistently.

3. Feature Engineering and Exploratory Structure

General and Sector-specific Accounting Features

The feature engineering process expanded the 403-column build into a 768-column analysis file. The new variables were designed to convert raw statements into economically interpretable ratios, growth measures, yields, and sector themes. General features included

profitability, return on assets and equity, liquidity, leverage and solvency, cash-flow quality, working-capital efficiency, per-share metrics, valuation multiples, quarter-over-quarter and year-over-year growth, accrual measures, and shareholder payout variables. The pipeline also built sector-specific variables for Utilities, Banks, Insurance, REITs, and Energy/Oil & Gas as generic industrial ratios do not fully represent the economics of these businesses.

Table 2

Representative Engineered Feature Families

Feature family	Examples used in the project
Profitability / returns	ROA, ROE, operating ROA, pretax ROA, gross and net margins
Liquidity / leverage	current ratio, cash ratio, debt/assets, debt/equity, net debt/assets
Cash flow / quality	FCF margin, OCF/net income, FCF/debt, accrual ratio
Valuation / per-share	P/B, P/S, P/OCF, P/FCF, FCF yield, cash/share
Growth	revenue, net income, assets, debt, OCF: QoQ and/or YoY
Utilities	regulatory assets, fuel cost, generation, revenue/customer
Banks	loan/deposit, capital ratios, NIM, efficiency, credit-loss metrics
Insurance / REIT / Energy	claims and premium ratios; FFO/AFFO; production/reserve/refining measures

Sector-specific fields that were structurally not applicable to a company were handled separately from ordinary missingness. This design avoids treating the absence of a bank only or utility only metric as the same type of missing value as an unavailable general accounting field.

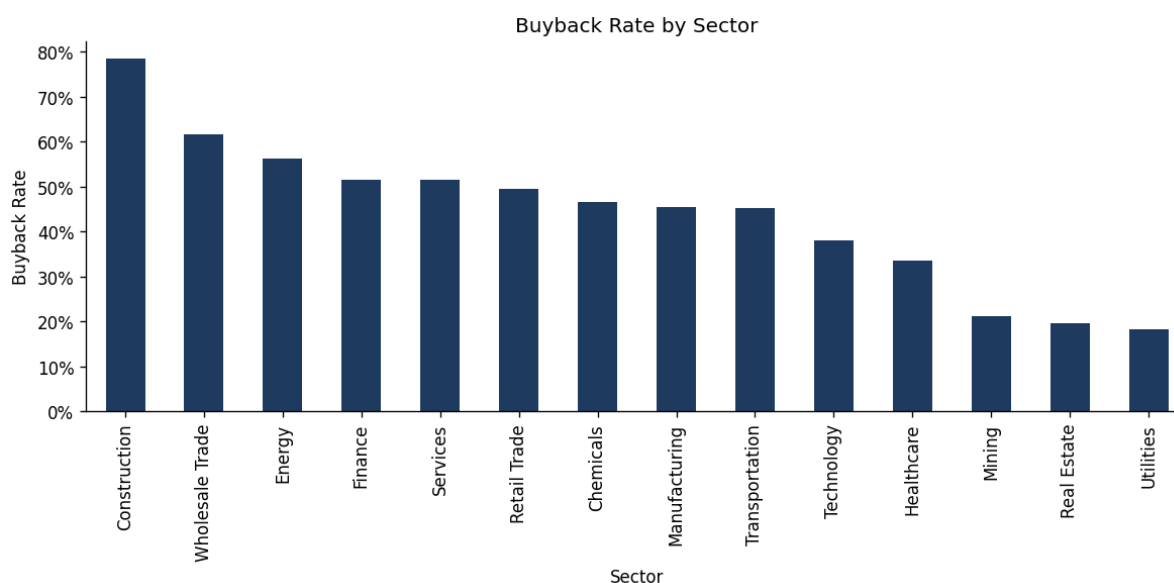
Exploratory Patterns

The final exploratory dataset contains 17,319 observations, 768 columns, 954 tickers, and all four fiscal-quarter labels. Interestingly, buyback prevalence is not constant across time or industry. The yearly rate ranges from roughly 39% to 47% over the displayed 2021-2025 period, while sector-level rates differ much more sharply. Construction and Wholesale Trade display substantially higher buyback frequencies than Utilities, Real Estate, and Mining in the sample.

These differences justify retaining sector and time controls rather than assuming one homogeneous repurchase process.

Figure 2

Observed Next-Quarter Buyback Rate by Sector



The exploratory analysis then revealed extreme skewness in many accounting ratios and valuation multiples. Financial ratios become unstable when denominators approach zero, producing long tails that can dominate distance-based or linear estimators. This observation directly informed the preprocessing strategy. Outlier capping, log transforms for safe non-negative scale variables, and sector-based median imputation were learned from the training data and then applied unchanged to validation and test data.

4. Modeling Design

Train, Validation, and Test Split

The 17,319 labeled observations were divided with a stratified random 60/20/20 split: 10,391 training rows, 3,464 validation rows, and 3,464 test rows. Each partition retained a 42.6%

positive-class rate. Of course, the test set was held aside for one final evaluation after model and threshold selection.

Train-only Preprocessing

Preprocessing was deliberately learned from the training sample to reduce leakage between model-development stages. The original predictor matrix contained 767 columns, and columns that were all missing, constant, or all zero in training were removed. General, sector-agnostic columns with more than 30% training sample missingness were also removed, while sector-specific engineered features were retained under their separate missingness logic. This reduced the shared pre-encoded feature set from 767 to 372 columns.

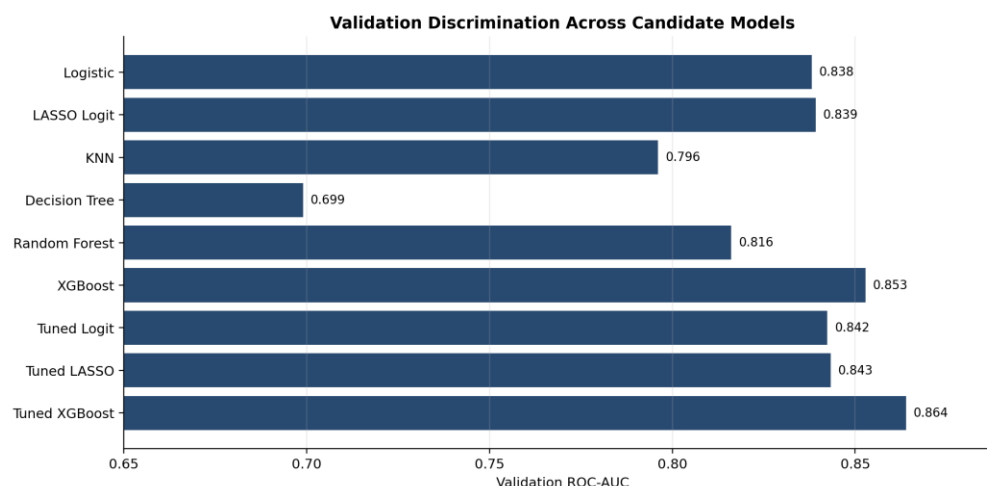
1. Outlier treatment: numeric engineered variables from the accrual-feature region onward were winsorized at the 1st and 99th percentiles using training-derived caps; validation and test values were clipped to those same caps.
2. Identifiers and non-model fields: ticker, company name, API ticker fields, price-date identifiers, exchange, and other designated fields were removed. Period-end date was transformed into fiscal month and then dropped.
3. Missing numeric values: sector-specific medians estimated on the training set were used first with the overall training median as fallback.
4. Scale transforms: $\log_{1p}$ was applied to selected non-negative scale variables such as revenue, market capitalization, total assets, liabilities, debt, and average balance-sheet levels.
5. Categorical encoding: fiscal period, fiscal month, statement currencies, sector, and industry were one-hot encoded; validation and test matrices were reindexed to the training columns.

6. Scaling: StandardScaler was used for distance/linear models where appropriate, while tree-based models were left on their natural feature scales.

After one-hot encoding, the tree-model design matrix used by the main XGBoost specification contained 654 inputs. An alternative fundamentals-only model removed direct buyback/share-change predictors and used 651 inputs.

Candidate algorithms and selection criterion

The project compared logistic regression, L1-regularized logistic regression, k-nearest neighbors, a single classification tree, random forest, and XGBoost. ROC-AUC was emphasized for model selection because it measures ranking quality across classification thresholds. Accuracy, sensitivity, specificity, and F1 were retained as complementary diagnostics. The main XGBoost model was tuned with five-fold cross-validation over 324 hyperparameter combinations (1,620 fits). The selected parameters were 300 estimators, maximum depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8, and zero L1 penalty. A validation threshold search from 0.30 to 0.70 selected 0.4600 by validation accuracy.

Figure 3*Validation ROC-AUC by Candidate Model*

5. Predictive Results and Interpretation

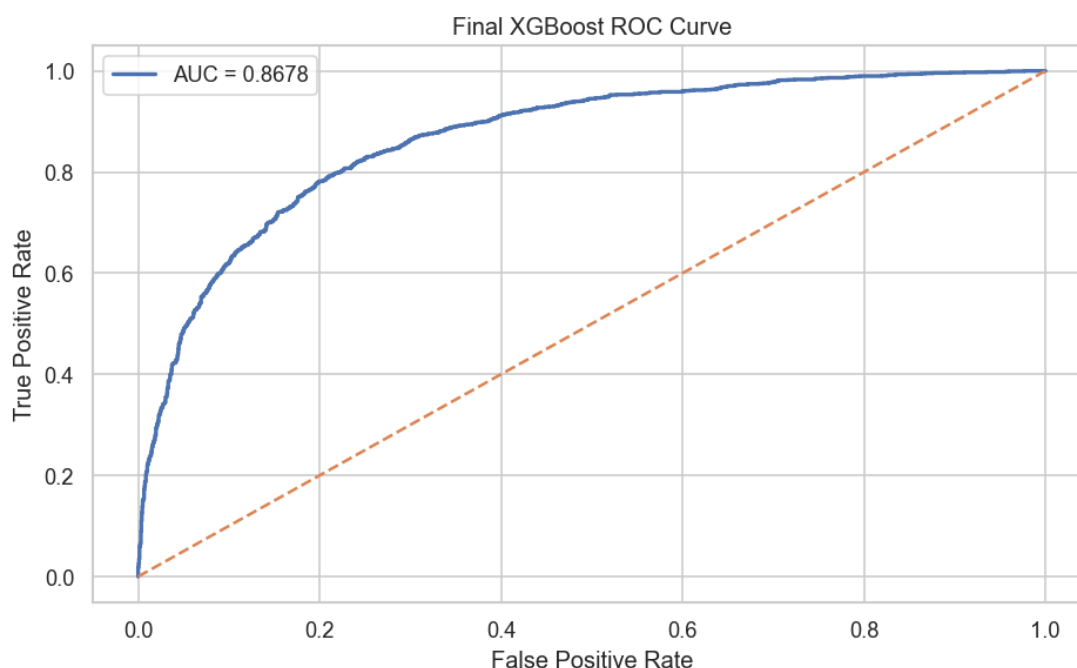
Final Held-Out Performance

At the selected 0.4600 threshold, the final main XGBoost model achieved 78.93% accuracy and a 0.8678 ROC-AUC on the held-out test set. It correctly identified 73.97% of actual next-quarter buybacks and 82.60% of non-buybacks with a 0.7493 F1 score. The test confusion matrix contains 1,643 true negatives, 346 false positives, 384 false negatives, and 1,091 true positives.

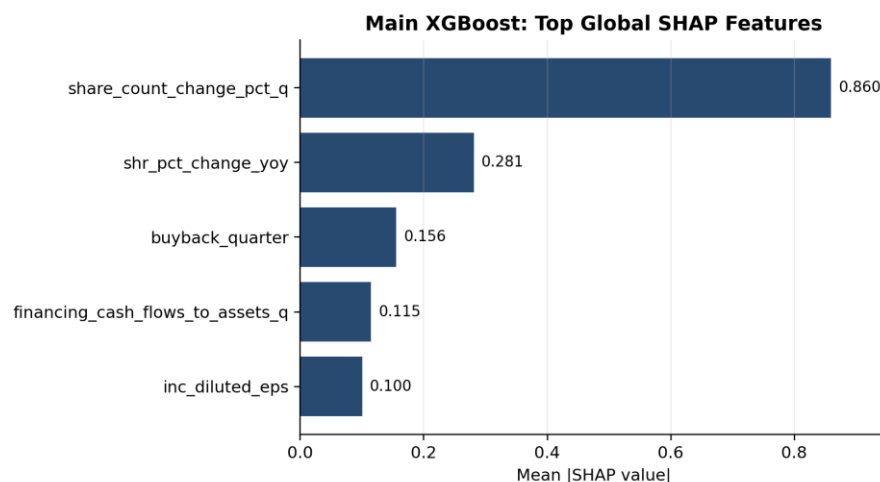
Table 3*Final Main XGBoost Performance*

Metric	Validation	Held-out test
Threshold	0.4600	0.4600
Accuracy	0.7965	0.7893
ROC-AUC	0.8639	0.8678
Sensitivity	0.7714	0.7397
Specificity	0.8151	0.8260

Metric	Validation	Held-out test
F1	0.7633	0.7493

Figure 4*Held-Out Test ROC Curve***What the Model Learned**

The strongest features in the main model are closely related to recent share-count dynamics. Mean absolute SHAP importance ranks current quarter share-count percentage change first, followed by year-over-year share-count change, the current-quarter buyback indicator, financing cash flows relative to assets, and diluted EPS. This is economically intuitive: corporate payout policy is persistent, and financing cash flows and share counts directly reflect the capital-allocation process as SHAP values provide a unified feature-attribution framework for interpreting complex model predictions.

Figure 5*Global SHAP Feature Importance*

That dominance creates an important diagnostic question: is the model simply forecasting persistence? A logistic baseline using only the current-quarter buyback indicator achieved 0.7763 validation AUC and 78.35% accuracy. The full model therefore adds substantial ranking information beyond that single persistence indicator even though the baseline accuracy is already relatively high because of the class structure and persistence in buyback behavior.

Fundamentals-Only Robustness Test

A second XGBoost specification removed the most direct buyback history and share-change variables of current-quarter buyback, prior period share change, quarterly and annual share growth fields, share-count change and percentage change, and year-over-year share-count change. Five-fold grid search selected 300 trees, depth 6, learning rate 0.05, subsample 0.7, column subsample 0.8, and L1 penalty 0.1. With a validation-selected threshold of 0.49, the held-out test AUC was 0.8611, test accuracy was 78.55%, and F1 was 0.7487.

Table 4*Robustness to Removing Direct Buyback and Share-Change Predictors*

Specification	Test AUC	Accuracy	Sensitivity	Specificity	F1
Main XGBoost	0.8678	0.7893	0.7397	0.8260	0.7493
Fundamentals-only XGBoost	0.8611	0.7855	0.7505	0.8115	0.7487

The decline in AUC is only 0.0067. This result supports a narrower claim than "fundamentals cause buybacks" to show that the dataset contains predictive information about future repurchase behavior outside the direct share-count persistence variables. In the alternative model, basic EPS, sector indicators, net financing cash flow, income before taxes, regulatory-assets ratios, tax-burden measures, and sector-specific variables all contribute to the classification.

6. From Buyback Prediction to an Investment Signal

Information Timing and Portfolio Formation

The portfolio stage uses the fitted main XGBoost model to generate a frozen buyback probability for every observation in the held-out test sample. The analysis then imposes an information availability rule before any return is computed. Because actual filing dates were not available in the project dataset, each accounting observation is assumed to become available 90 days after its fiscal period end. The quarter containing that information date is the signal quarter; the portfolio is formed at the beginning of the following calendar quarter and exits at the beginning of the next quarter. This creates an explicit lag between the accounting period end and investment action.

Daily adjusted prices were downloaded for the required backtest tickers from Yahoo Finance. For each common quarter start, the first actual trading day on or after the target date

was used for every stock in that quarter. The frozen price cutoff was August 10, 2026. Of 929 requested tickers, 907 downloaded successfully; 3,356 of the 3,449 candidate observations had valid future returns, leaving 93 without a complete mapped return.

Cross-Sectional Signals

The stock-selection experiment compares the model probability with three conventional characteristics already available or constructible from the project data. Value is free-cash-flow yield (FCF_yield_q). Profitability is ROA, leverage is debt-to-assets, and Quality is constructed as the average of high profitability and low leverage after within-quarter standardization. Momentum is a three-quarter compounded return drawn from approximately four, three, and two quarters before formation, intentionally skipping the most recent quarter. The buyback signal is the XGBoost probability calculated before. These signal choices follow the broad cross-sectional asset-pricing standards represented by Fama and French and Mark Carhart.

Value, ROA, leverage, and Momentum are winsorized within formation quarter at the 1st and 99th percentiles and converted to within-quarter z-scores. Buyback probability is also standardized by formation quarter. Quality is computed as $(z(\text{ROA}) - z(\text{leverage}))/2$ and then standardized again. The baseline VMQ score is the equal-weight average of standardized Value, Momentum, and Quality. VMQ+BB adds the standardized buyback probability as a fourth equally weighted component.

$$\text{VMQ} = (z\text{Value} + z\text{Momentum} + z\text{Quality}) / 3$$

$$\text{VMQ+BB} = (z\text{Value} + z\text{Momentum} + z\text{Quality} + z\text{Buyback}) / 4$$

Requiring complete future return and all four standardized signals yielded 2,073 eligible observations. A minimum size rule retained only quarters where a Top 10% portfolio would contain at least ten stocks, producing a final portfolio sample of 1,998 stock-quarters across 15

formation quarters from 2022Q3 through 2026Q1. The included quarterly universe ranged from 111 to 154 stocks.

Portfolio Construction, Turnover, and Costs

For Buyback, VMQ, and VMQ+BB, equal-weight Top 10% and Top 20% portfolios were formed each quarter. An equal-weight portfolio of all eligible names served as the benchmark. Turnover was calculated as one-half the absolute change between new target weights and prior-quarter post-return drifted weights (the initial portfolio purchase was treated as 100% turnover). Net returns were evaluated under 10, 25, and 50 basis point transaction cost assumptions per unit of turnover. Performance statistics include annualized compounded return, annualized volatility, a zero risk-free rate Sharpe ratio, maximum drawdown, and hit rate relative to the equal-weight benchmark.

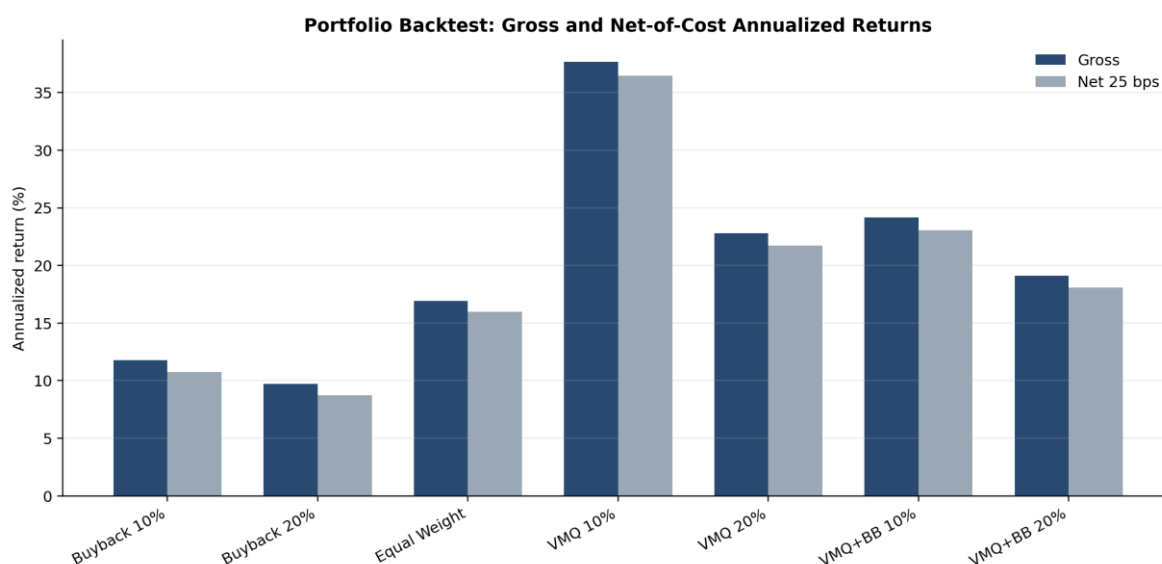
7. Portfolio Results

Performance

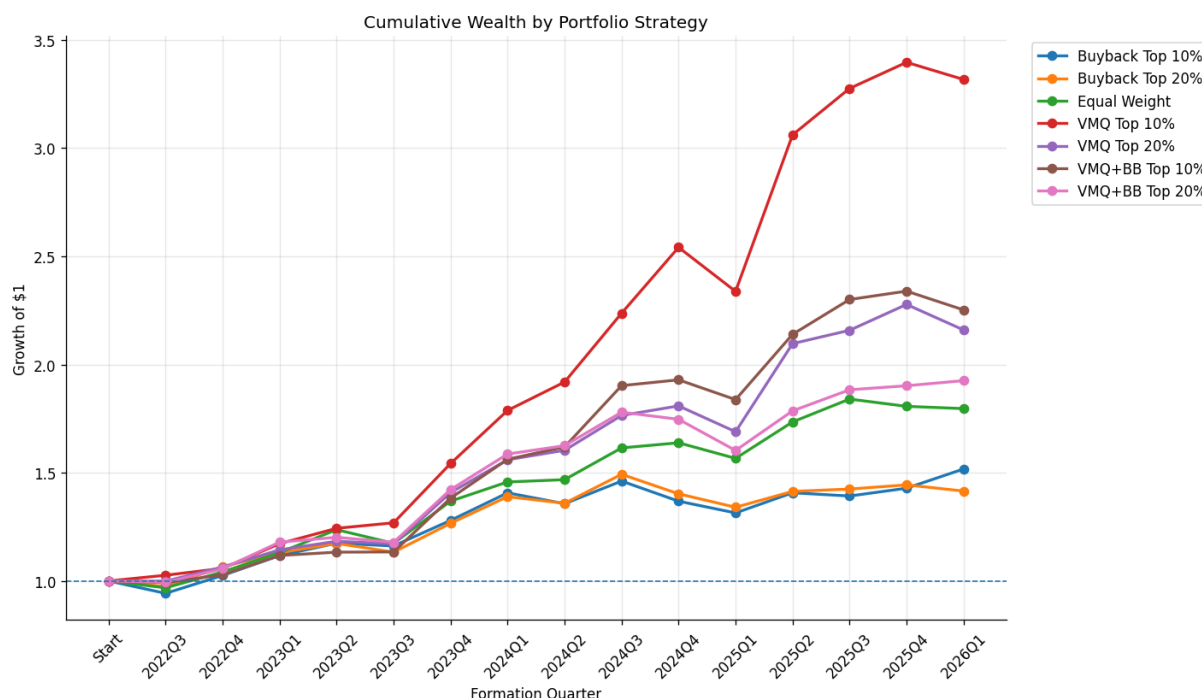
Unfortunately, the buyback probability was not a strong standalone return-ranking signal. The Buyback Top 10% portfolio returned 11.76% annualized gross and 10.74% after 25 basis points of turnover costs, versus 16.91% and 15.97% for the equal-weight benchmark. The Buyback Top 20% portfolio was weaker still at 9.71% gross annualized. In contrast, the Value-Momentum-Quality composite produced the strongest results: VMQ Top 10% returned 37.68% gross annualized with a 1.78 Sharpe ratio, while VMQ Top 20% returned 22.79%.

Table 5*Backtest Performance, 2022 Q3-2026 Q1 (15 Quarters)*

Strategy	Gross ann.	Net 25 bps	Ann. vol.	Sharpe	Max DD
Buyback Top 10%	11.76%	10.74%	12.08%	0.99	-10.07%
Buyback Top 20%	9.71%	8.74%	11.84%	0.85	-10.19%
Equal Weight	16.91%	15.97%	12.97%	1.29	-5.08%
VMQ Top 10%	37.68%	36.49%	19.66%	1.78	-8.01%
VMQ Top 20%	22.79%	21.74%	16.71%	1.33	-6.60%
VMQ+BB Top 10%	24.18%	23.06%	16.16%	1.45	-4.74%
VMQ+BB Top 20%	19.09%	18.06%	14.56%	1.29	-9.93%

Figure 6*Annualized Portfolio Returns Before and After Transaction Costs*

As expected, the signals were quite expensive to rebalance. Average quarterly turnover was 93.58% for Buyback Top 10%, 93.30% for VMQ Top 10%, and 94.78% for VMQ+BB Top 10%; Top 20% portfolios were near 90.5%-90.7%, and even the eligible universe equal-weight benchmark turned over 83.45% on average. The project therefore reports cost sensitivity rather than presenting gross returns alone.

Figure 7*Growth of One Dollar by Strategy***Cross-Sectional Information Coefficients**

Quarterly Spearman rank information coefficients provide a ranking level production independent of an arbitrary portfolio cutoff. The buyback signal had a mean IC of 0.0061, a median IC of -0.0123, and a positive IC in 46.7% of quarters. VMQ and VMQ+BB were also weak on average across all names with mean ICs of 0.0127 and 0.0112 respectively. In an extended signal comparison, Quality produced the clearest positive average rank relationship (mean IC approximately 0.054, positive in 73.3% of quarters), while Value was modestly positive and Momentum was near zero.

Nevertheless, the buyback score was not simply duplicating Momentum. Average signal correlations were about 0.31 with Value, 0.08 with Momentum, and 0.28 with Quality. Thus, the

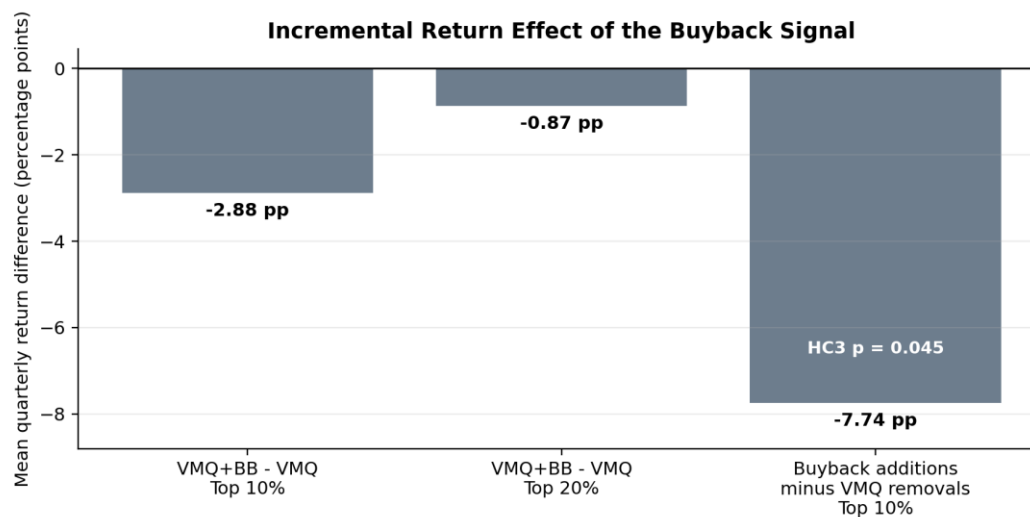
lack of incremental performance cannot be explained by perfect redundancy. The signal changed portfolio composition, but those changes were generally unfavorable in the realized sample.

Does Buyback Probability improve VMQ?

This is the most direct investment test in the project. If the predicted buyback probability contains useful incremental return information, an equally weighted VMQ+BB score should improve upon the same VMQ ranking. However, it simply failed to do so. At the Top 10% cutoff, VMQ+BB returned 24.18% gross annualized versus 37.68% for VMQ. The average quarter-by-quarter difference was -2.88 percentage points and was positive in only 33.3% of quarters. At the Top 20% cutoff, the average difference was -0.87 percentage points and was also positive in only one-third of quarters.

Figure 8

Incremental Quarterly Return Effects of Adding the Buyback Signal



Note. The buyback score reduced the VMQ composite on average. The replacement-level Top 10% test is statistically significant at the 5% level using HC3 robust inference.

Replacement analysis: why the Top 10% composite weakened

The Top 10% VMQ and VMQ+BB portfolios shared 65.9% of names on average, making it possible to isolate the actual substitutions induced by adding the buyback signal. Stocks removed from the pure VMQ Top 10% ranking subsequently earned 9.88% per quarter on average, while stocks added because of the buyback score earned 2.14%. The average added-minus-removed spread was therefore -7.74 percentage points, with a median of -6.01 percentage points. Buyback additions underperformed the names they displaced in 66.7% of quarters. An intercept-only OLS test with HC3 robust standard errors produced $t = -2.20 / p = 0.045$ with a 95% confidence interval of approximately -15.28 to -0.21 percentage points.

The buyback signal was sufficiently different from VMQ to change the selected stocks, but in this 15-quarter sample, it tended to promote stocks with weaker subsequent returns. The negative result is therefore more informative than the statement that buyback and VMQ were redundant.

8. Factor-Adjusted Returns

To separate raw portfolio returns from common factor exposures, monthly Fama-French five factor and Momentum data were downloaded through pandas-datareader, compounded to calendar-quarter returns, and merged to every backtest quarter. Excess portfolio returns were regressed on CAPM, Fama-French three factor, Carhart four factor, and Fama-French five factor plus Momentum specifications. OLS regressions used HC3 robust standard errors and t-based inference because only 15 quarterly observations were available per strategy.

Table 6*Fama-French Five-Factor Plus Momentum Regressions (HC3)*

Strategy	FF5+Mom alpha/qtr	Annualized alpha	t-stat	p-value	R-squared
Buyback Top 10%	0.80%	3.26%	0.24	0.818	0.569
Buyback Top 20%	-0.85%	-3.34%	-0.40	0.698	0.691
Equal Weight	0.61%	2.46%	0.47	0.650	0.890
VMQ Top 10%	5.47%	23.74%	1.85	0.101	0.791
VMQ Top 20%	2.23%	9.22%	0.90	0.396	0.803
VMQ+BB Top 10%	1.24%	5.06%	0.56	0.592	0.718
VMQ+BB Top 20%	1.00%	4.04%	0.31	0.762	0.734

Unfortunately, none of the strategy alphas is statistically significant at the 5% level.

VMQ Top 10% has a large economic alpha estimate but only $t = 1.85/p = 0.101$, illustrating the uncertainty created by the short sample. More directly, the VMQ+BB-minus-VMQ Top 10% return spread has negative estimated alpha across all four factor specifications. The Carhart four-factor version is -3.11 percentage points per quarter ($p = 0.034$); the FF5+Momentum version is -4.23 percentage points ($p = 0.089$). At the Top 20% cutoff the difference estimates are smaller and clearly imprecise.

These regressions are best interpreted as robustness diagnostics, not definitive asset pricing tests. Fifteen quarterly observations are insufficient for a strong claim about long run alpha, particularly when six factors plus an intercept are estimated in this specification. The limited quarterly observations are in part due to the limited dataset available to an undergraduate student.

9. Limitations

The project is intentionally transparent about constraints that could affect both classification performance and investment conclusions. Several limitations are material enough that a reproduction of this project should address them directly.

1. Universe and survivorship: The starting file is a Russell 3000 constituent list used at the time of the project, not a point-in-time historical constituent database. Delisted firms and historical membership changes are not reconstructed.
2. API coverage: Only 955 of 2,585 attempted tickers produced firm-quarter panels. The resulting 954-issuer labeled sample is therefore a coverage subset of the intended universe, not the complete Russell 3000.
3. Random row split: Train, validation, and test observations are stratified randomly by row. Because a firm appears in multiple quarters, the same issuer can occur in multiple partitions. This is appropriate for the project as implemented but can overstate generalization to unseen issuers and does not test temporal regime changes.
4. Label interpretation: The hybrid label uses a provider flag or a share count decline of at least 1.25%. A declining share count is an observable proxy for net repurchase activity but can be affected by issuance, compensation, corporate actions, and data differences.
5. Information date approximation: The portfolio stage lacks actual filing timestamps and therefore uses a 90-day lag from period end. This is an explicit anti-look-ahead rule but not a substitute for exact 10-Q/10-K filing dates.
6. Complete case backtest contraction: The application stage falls from 3,449 candidate observations to 2,073 complete-signal observations then 1,998

observations across 15 valid formation quarters. Missing Momentum, factors, and prices materially narrow the investable sample.

7. Price availability: Yahoo Finance returned usable daily data for 907 of 929 requested tickers; 93 candidate observations ultimately lacked a valid future return.
8. Short backtest window: The final comparison covers 2022Q3-2026Q1. Annualized returns, Sharpe ratios, and factor alphas are sensitive to this market regime and to the small number of quarters.
9. Trading friction: The cost model applies simple basis-point costs to turnover and does not explicitly model bid-ask spread, market impact, taxes, liquidity constraints, etc.
10. Signal/application consistency: The portfolio study uses the main XGBoost probability, which includes recent buyback/share-change history. The fundamentals-only model is a classification robustness test. Its probabilities were not separately backtested in the submitted application notebook.

10. Conclusion

This project began with a prediction question and ended with a more useful distinction. Next quarter corporate share repurchases are predictable in the assembled firm quarter dataset. A tuned XGBoost classifier reached 0.8678 held-out ROC-AUC, and a fundamentals-only specification retained 0.8611 AUC after direct repurchase history variables were removed.

The portfolio evidence, however, does not support treating the probability as a standalone equity factor in the tested period. Buyback-only Top 10% and Top 20% portfolios

underperformed the equal-weight eligible universe. More importantly, adding the buyback probability to an equal-weight Value-Momentum-Quality composite reduced both Top 10% and Top 20% returns. The Top 10% replacement analysis shows a concrete mechanism: the buyback overlay displaced stocks that subsequently performed better, generating an average -7.74 percentage point added-minus-removed spread.

The research conclusion is that forecasting a corporate decision and forecasting the return to owning the company are different statistical tasks. The buyback model appears useful as a corporate behavior forecast and potentially as a conditional feature in broader decision systems, but this backtest does not establish it as a positive cross-sectional return signal. The next iteration should use point-in-time index membership, actual filing timestamps, grouped and time-based out-of-sample validation, a longer historical price horizon, and a separate backtest of the fundamentals-only probability. Those changes would test whether the negative investment result is structural or sample-specific while preserving the strongest part of the current work of an auditable pipeline from raw financial data to model interpretation and investable hypotheses.

Data Sources and Reproducibility

The analysis was implemented in Python/Jupyter. The notebooks document the complete workflow across dataset construction, feature preparation, exploratory analysis, model assembly, and portfolio application. Primary data interfaces used in the project were the Russell 3000 constituent file used to create the universe; StockAI Insights for quarterly financial statements, share data, and company profiles; Yahoo Finance through `yfinance` for historical prices; and the Kenneth R. French Data Library through `pandas-datareader` (French) for Fama-French and Momentum returns. Core modeling and inference libraries included `pandas`, `NumPy`, `scikit-learn`, `XGBoost`, `SHAP`, `SciPy`, and `statsmodels`.

Implementation note: selected helper functions, visualization syntax, and debugging steps were AI-assisted and are identified as such in the underlying notebooks. Research framing, feature choices, model comparisons, portfolio definitions, and interpretation are presented here as the author's project decisions.

References

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Appendix A

Selected Model Specifications

Table A1

Candidate and Final Model Configurations

Model	Key specification / selected parameters
Logistic regression	Standardized continuous inputs; one-hot categorical inputs
Tuned logistic	L1 penalty; C=1; balanced class weights; liblinear; 5-fold AUC grid search
Tuned LASSO logistic	L1 penalty; C=0.1; balanced class weights; threshold 0.59
KNN	k=3; scaled non-dummy inputs
Decision tree	Gini criterion; baseline unpruned tree used for comparison
Random forest	500 trees; max_features=4; min_samples_leaf=5; max_leaf_nodes=30; OOB score 0.7489
Main XGBoost	300 trees; depth 6; eta 0.05; subsample 0.8; colsample 0.8; reg_alpha 0; threshold 0.46
Fundamentals-only XGBoost	300 trees; depth 6; eta 0.05; subsample 0.7; colsample 0.8; reg_alpha 0.1; threshold 0.49

Appendix B

Transaction-Cost Sensitivity

Table B1

Annualized Return Under Alternative Transaction-Cost Assumptions

Strategy	Gross	Net 10 bps	Net 25 bps	Net 50 bps
Buyback Top 10%	11.76%	11.35%	10.74%	9.73%
Buyback Top 20%	9.71%	9.32%	8.74%	7.77%
Equal Weight	16.91%	16.53%	15.97%	15.03%
VMQ Top 10%	37.68%	37.20%	36.49%	35.31%
VMQ Top 20%	22.79%	22.37%	21.74%	20.69%
VMQ+BB Top 10%	24.18%	23.73%	23.06%	21.96%
VMQ+BB Top 20%	19.09%	18.68%	18.06%	17.04%

Appendix C

Key Incremental Tests

Table C1

Final Inference Summary Using HC3 Robust Standard Errors

Test	Estimate / quarter	t-stat	p-value	95% CI
Buyback Top 10% FF5+Momentum alpha	0.80%	0.24	0.818	[-6.99%, 8.60%]
Buyback Top 20% FF5+Momentum alpha	-0.85%	-0.40	0.698	[-5.71%, 4.01%]
VMQ+BB - VMQ Top 10%, FF5+Mom alpha	-4.23%	-1.93	0.089	[-8.51%, 0.06%]
VMQ+BB - VMQ Top 20%, FF5+Mom alpha	-1.23%	-0.29	0.783	[-9.70%, 7.23%]
BB-added - VMQ-removed names, Top 10%	-7.74%	-2.20	0.045	[-15.28%, -0.21%]

Appendix D

Project File Sequence

1. Building the Dataset - Russell 3000 cleaning, API retrieval, panel assembly, and buyback label.
2. Preparing the Dataset - broad and sector-specific financial feature engineering.
3. Exploring the Data - class balance, time/sector patterns, distributions, relationships, and market-cap analysis.
4. Assembling the Models - train-only preprocessing, model comparison, tuning, SHAP interpretation, persistence baseline, and fundamentals-only robustness model.
5. Applying the Models - frozen probabilities, information lags, price mapping, factor construction, portfolio formation, costs, information coefficients, factor regressions, and replacement analysis.